Class 4

Mathematics

Solutions

Olympiad Program

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1. Number sense

Solutions

Worksheet – 1

1. **Correct Answer: (A) One lakh twenty-three thousand four hundred fifty-six**
   1,23,456
   = 1 lakh + 23 thousand + 456
   Correct Answer: (A) One lakh twenty-three thousand four hundred fifty-six
2. **Correct Answer: (A) One lakh five thousand two hundred thirty 1,05,230**
   = 1 lakh + 5 thousand + 230
3. **Correct Answer: (A) Five lakh eight thousand nine hundred twelve**
   5,08,912
   = 5 lakh + 8 thousand + 912
4. **Correct Answer: (B) Two lakh thirty-four thousand five hundred sixty-seven**
   2,34,567
   = 2 lakh + 34 thousand + 567
5. **Correct Answer: (B) Fifty-six thousand seven hundred nine**
   56,709
   = 56 thousand + 709
6. **Correct Answer: (B) 92,581**
   Ninety-two thousand five hundred eighty-one
   = 92,000 + 500 + 81
7. **Correct Answer: (A) Four lakh twenty-eight thousand three hundred fifteen**
   4,28,315
   = 4 lakh + 28 thousand + 315

Worksheet – 2

1. **Correct Answer: (B)**
   Place value of 5 in 52,347
   5 is in ten-thousands place ⟶ 5 × 10,000 = 50,000
2. **Correct Answer: (A)**
   Face value of 3 in 63,215
   Face value = digit itself ⟶ 3
3. **Correct Answer: (D)**
   Place value of 2 in 1,25,430
   2 is in ten-thousands place ⟶ 2 × 10,000 = 20,000
4. **Correct Answer: (A)**
   Sum of place values of 6 in 76,245
   6 is in thousands place ⟶ 6 × 1,000 = 6,000
5. **Correct Answer: (A)**
   Digit with place value 30,000
   30,000 = 3 × 10,000 ⟶ digit is 3
6. **Correct Answer: (C)**
   In 5,46,218
   - Face value of 5 = 5
   - 5 is in lakh place ⟶ 5 × 1,00,000 = 5,00,000
7. **Correct Answer: (B)**
   Number = 25,468
   - Thousands place = 5
   - Tens place = 6
     After interchange ⟶ 26,458

   Compare:25,468 < 26,458
   Original number is smaller than new number

Worksheet – 3

1. **Correct Answer: (B)**
   4,302
   = 4,000 + 300 + 2
2. **Correct Answer: (A)**
   2,000 + 500 + 40 + 8
   = 2,548
3. **Correct Answer: (B)**
   6,709
   = 6,000 + 700 + 9
4. **Correct Answer: (A)**
   8,000 + 600 + 50
   = 8,650
5. **Correct Answer: (A)**
   3,045
   = 3,000 + 40 + 5

6. **Correct Answer: (A)**
   9,824
   = 9,000 + 800 + 20 + 4
   Missing value = 800
7. **Correct Answer: (A)**
   4,320
   = 4,000 + 300 + 20 + 0
   Missing value = 20

Worksheet – 4

1. **Correct Answer: (A)**
   Predecessor of 10,001
   = 10,001 – 1 = 10,000
2. **Correct Answer: (A)**
   Successor of 99,999
   = 99,999 + 1 = 100,000
3. **Correct Answer: (D) 5946**

   Order: Greatest to least
   Given: 7952 > 5642 >? > 2341
   So missing number must be between 5642 and 2341

   Check options:

   - 2431 (fits)
   - 3241 (fits)
   - 5611 (fits)
   - 5946 (greater than 5642 → wrong position)
4. **Correct Answer: (A)**

   Round 7,843 to nearest 10
   Last digit = 3 (<5) ⟶ round down
   = 7,840
5. **Correct Answer: (B)**

   Round 92,582 to nearest hundreds
   Last two digits = 82 (⩾50) ⟶ round up
   = 92,600
6. **Correct Answer: (B)**

   Ascending order (smallest ⟶ greatest)
   Correct order: 12,234 < 12,345 < 12,543
7. **Correct Answer: (C) P is the smallest among all the numbers**

   Given numbers:
   P = 2709, Q = 2799, R = 2750, S = 2790

   Smallest = 2709 (P)
   Largest = 2799 (Q)

Worksheet – 5

1. **Correct Answer: (A)**
   Smallest 4-digit number (2, 8, 1, 9)
   Arrange in ascending order ⟶ 1, 2, 8, 9 ⟶ 1289
2. **Correct Answer: (C)**
   Largest number (3, 6, 1)
   Arrange in descending order ⟶ 6, 3, 1 ⟶ 631
3. **Correct Answer: (A)**
   Smallest 4-digit number (8, 0, 4, 5)
   First digit cannot be 0 ⟶ smallest non-zero = 4
   Then arrange ⟶ 4, 0, 5, 8 ⟶ 4058
4. **Correct Answer: (B)**
   Roman numeral for 90
   = XC
5. **Correct Answer: (C)**
   Largest Roman numeral
   IX = 9, XL = 40, XC = 90, LXX = 70
   Largest = 90 (XC)
6. **Correct Answer: (A)**
   VI × V
   6 × 5 = 30 ⟶ XXX
7. **Correct Answer: (B)**
   L + XX
   50 + 20 = 70 ⟶ LXX

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (A)**
   53,284
   = 53 thousand + 284
   ⟶ Fifty-three thousand two hundred eighty-four
2. **Correct Answer: (A)**
   Three lakh twelve thousand six hundred forty-five
   = 3,00,000 + 12,000 + 645
   = 3,12,645
3. **Correct Answer: (B)**
   Descending order (greatest ⟶ smallest)
   P = 38750, Q = 38075, R = 38705, S = 38775
   Order: 38775 > 38750 > 38705 > 38075
   ⟶ S, P, R, Q

4. **Correct Answer: (B)**
   50 + 40 + 3
   50 = L, 40 = XL, 3 = III
   → L + XL + III = LXLIII
5. **Correct Answer: (C) 5373**
   Greatest number = 7420 (arrange in descending order)
   Smallest number = 2047 (smallest digit cannot be 0 in first place → use 2 first)
   Difference = 7420 – 2047 = 5373
6. **Correct Answer: (D) 900000**
   Greatest 6-digit number = 999999
   Greatest 5-digit number = 99999
   Difference = 999999 – 99999 = 900000
7. **Correct Answer: (C) 66951**
   Greatest number = 97630 (descending order)
   Smallest number = 30679 (0 cannot be first, so start with 3)
   Difference = 97630 – 30679 = 66951
8. **Correct Answer: (B) Two lakh seventy six thousand fifteen**
   2,76,015 = Two lakh seventy six thousand fifteen
9. **Correct Answer: (D) 30459**
   Smallest 5-digit number:
   - First digit cannot be 0, so take smallest non-zero → 3
   - Arrange remaining digits in ascending order → 0, 4, 5, 9

   Number = 30459
10. **Correct Answer: (A) 32945**
    Thirty two thousand = 32,000
    Nine hundred forty five = 945
    Number = 32,945
11. **Correct Answer: (A) Only P is true**
    P: 584921 → nearest thousand = 585000
    - Q: Place value of 2 = 20000, 5 = 5000 → sum = 25000, not 26000

12. **Correct Answer: (A) 2510**
    To get the greatest number, arrange digits in descending order:
    5, 2, 1, 0 → 5210
    But this option is not given, so check closest valid option:
    2510 is the greatest among given choices
13. **Correct Answer: (C) R**
    P: CLX = 100 + 50 + 10 = 160
    - Q: DCCXL = 500 + 200 + 40 = 740
    - R: CMLXIX = 900 + 60 + 9 = 969

    Comparing: 969 > 740 > 160
14. **Correct Answer: (B) 3578**
    2790 → 7 is in hundreds place
    - 3578 → 7 is in tens place
    - 4789 → 7 is in hundreds place
    - 5785 → 7 is in hundreds place

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answers: (A), (B)**
    Expanded forms of 84,605
    = 80,000 + 4,000 + 600 + 5
    (A) correct
    (B) correct (same in multiplication form)
    (C) missing 4,000
    (D) missing 600
16. **Correct Answers: (A), (C)**
    Digit 4 in ten-thousands place
    Check first digit:
    (A) 40,521
    (B) 54,210
    (C) 46,789
    (D) 14,023
17. **Correct Answers: (A), (C)**
    Between 52,000 and 53,000
    52,500
    52,900
    51,999 (too small)
    53,100 (too large)
18. **Correct Answers: (A), (C), (D)**
    Correct Roman numerals
    XL = 40
    XC = 90
    LXX = 70 (not 60)
    CXL = 140
19. **Correct Answers: (A), (C), (D)**
    Place value statements
    (A) 7 in 70,000 → 70,000
    (B) 2 in 20,456 → 20,000 (not 2,000)
    (C) 3 in 13,245 → 3,000
    (D) 4 in 4,56,123 → 4,00,000

ASSERTION AND REASON TYPE

20. **Correct Answer: (A**

    In XL, X (10) is before L (50), so we subtract → 50 – 10 = 40.
    Thus, the reason correctly explains the assertion.

21. **Correct Answer: (D)Assertion is false, Reason is true.**

    Assertion (A): True → In 56,712, digit 5 is in ten-thousands place → value = 50,000 (not 5,000)
    So Assertion is false.

    Reason (R): True → Place value depends on position

22. **Correct Answer: (A)Both are true and Reason correctly explains Assertion.**

    Assertion (A): True → 3,52,607 > 3,25,760

    Reason (R): True → We compare from leftmost digit

MARTIX MATCHING TYPE

23. **Correct Answers: (a–Q), (b–P), (c–S), (d–R)**
    - Forty-five thousand, six hundred twelve → 45,612 → (Q)
    - Seventy-eight thousand, ninety → 78,090 → (P)
    - Sixty-five thousand, one hundred two → 65,102 → (S)
    - Eighty-three thousand, seven hundred → 83,700 → (R)
24. **Correct Answers: (a–Q), (b–P), (c–R), (d–S)**
    - XXV = 25 → (Q)
    - XLII = 42 → (P)
    - LXV = 65 → (R)
    - CXX = 120 → (S)

COMPREHENSION TYPE

Passage 1:

25. **Correct Answers: (A) ₹24,000**

    ₹24,350 → nearest thousand
    Since 350 < 500, round down

26. **Correct Answers: (A) ₱35,000**

    Rounded prices:
    - Dress → ₹24,000
    - Shoes (₹4,520 → ₹5,000)
    - Purse (₹6,200 → ₹6,000)

    Total = 24,000 + 5,000 + 6,000 = ₹35,000

27. **Correct Answers: (C) 500**

    In ₹4,520, digit 5 is in hundreds place
    Place value = 5 × 100 = 500

Passage 2:

28. **Correct Answers: (A) 478**

    Predecessor means one less than the number
    479 – 1 = 478

29. **Correct Answers: (A) Four hundred seventy-nine**

    479 in words = Four hundred seventy-nine

30. **Correct Answers: (C) 479**

    Successor means one more than the number
    478 + 1 = 479

2. Basic operations

Solutions

Worksheet – 1

1. **Correct Answer: (A) 77,889**
   $$45,678 + 32,211 = 77,889$$
2. **Correct Answer: Answer: (A) 80,110**
   $$56,432 + 23,678 = 80,110$$
   (Commutative property: order doesn't matter)
3. **Correct Answer: (B) ₹1,00,000**
   $$56,780 + 43,220 = 1,00,000$$
4. **Correct Answer: (A) 1,41,000**
   Round:
   $78,456 \approx 78,000,\ 62,543 \approx 63,000$
   Sum = $78,000 + 63,000 = 141,000$
5. **Correct Answer: (D) 4**
   $$65,4_3 + 23,145 = 88,588$$
   Check: $65,443 + 23,145 = 88,588$
   Missing digit = 4
6. **Correct Answer: (A) 60,601**
   $$24,356 + 36,245 = 60,601$$
7. **Correct Answer: (A) ₹70,912**
   $$24,567 + 46,345 = 70,912$$

Worksheet-2

1. **Correct Answer: (C) 11,111**
   $$92,345 - 81,234 = 11,111$$
2. **Correct Answer: (A) 23,000**
   Round:
   $45,678 \approx 46,000,\ 23,456 \approx 23,000$
   Difference = $46,000 - 23,000 = 23,000$
3. **Correct Answer: (A) 210,834**
   $$845,623 - 634,789 = 210,834$$
4. **Correct Answer: (A) Commutative Property**
   Order matters in subtraction → violates commutative property
5. **Correct Answer: (A) 33,224**
   $$78,902 - x = 45,678$$
   $$x = 33,224$$
6. **Correct Answer: (A) 32,667**
   $$56,234 - 23,567 = 32,667$$
7. **Correct Answer: (A) 140,000**
   $389,765 \approx 390,000,\ 245,421 \approx 250,000$
   $$390,000 - 250,000 = 140,000$$
8. **Correct Answer: (A) ₹23,298**
   $$76,543 - 53,245 = 23,298$$

Worksheet – 3

1. **Correct Answer: (A) 4,936**
   $$1,234 \times 4 = 4,936$$
2. **Correct Answer: (C) Identity Property**
   $23 \times 1 = 23$→ Identity property
3. **Correct Answer: (B) 31,000**
   $623 \approx 620,\ 48 \approx 50$
   $620 \times 50 = 31,000$
4. **Correct Answer: (C) 3,042**
   $234 \times 13 = 3,042$
5. **Correct Answer: (A) 14,940**
   $$1,245 \times 12 = 14,940$$
6. **Correct Answer: (B) 1,960,000**
   $4,897 \approx 4,900,\ 372 \approx 400$
   $4,900 \times 400 = 19,60,000 = 1,960,000$
7. **Correct Answer: (A) 108,864**
   $$4,536 \times 24 = 108,864$$

Worksheet – 4

8. **Correct Answer: (A) 5**
   1. 35 ÷ 7 = 5
9. **Correct Answer: (B) 49 days**
   245 ÷ 5 = 49
10. **Correct Answer: (C) 100**
    6,480 ÷ 64 ≈ 6,400 ÷ 64 = 100
11. **Correct Answer: (A) 80**
    12,000 ÷ 150 = 80
12. **Correct Answer: (B) 100**
    5. 2,500 ÷ 25 = 100
13. **Correct Answer: (B) 20**
    5,000 ÷ 250 = 20
14. **Correct Answer: (B) 2500**
    75,000 ÷ 30 = 2,500

Worksheet – 5

15. Correct Answer: (B) 180 km

60 × 3 = 180 km

16. Correct Answer: (A) $12

Cost of 1 book = 48 ÷ 12 = 4
Cost of 3 books = 4 × 3 = 12

17. Correct Answer: (A) 2 hours

More workers → less time
1 worker = 8 hours → 4 workers = 8 ÷ 4 = 2 hours

18. Correct Answer: (B) 7.5 cups

12 cookies → 3 cups
1 cookie → 3 ÷ 12 = 0.25 cup
30 cookies → 30 × 0.25 = 7.5 cups

19. Correct Answer: (A) $10

Cost of 1 pencil = 3 ÷ 15 = 0.2
Cost of 50 pencils = 50 × 0.2 = 10

20. Correct Answer: A) 12.5 litres

100 km → 5 L
1 km → 5/100 = 0.05 L
250 km → 250 × 0.05 = 12.5 L

21. Correct Answer: (A) 300 toys

120 toys in 2 hours → 60 toys/hour
5 hours → 60 × 5 = 300 toys

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. Correct Answer: (A) ₹47,177

95,412 – 48,235 = 47,177

2. Correct Answer: (B) 35,300

84,725 ≈ 84,700
49,381 ≈ 49,400
Difference ≈ 35,300

3. Correct Answer: D. 35

Each bike has 2 wheels → $27 \times 2 =$ 54wheels

- Remaining wheels → $194 - 54 = 140$
- Each car has 4 wheels → $140 \div 4 =$ 35cars

4. Correct Answer: B. CD + LXI = 665

A: XL = 40, CVI = 106 → $40 + 106 = 146$ Correct

B: CD = 400, LXI = 61 → $400 + 61 = 461 \neq 665$ Incorrect

C: M = 1000, DCL = 650 → $1000 - 650 = 350$ Correct

5. Correct Answer: A. 41742, 74147

Find **P**:

$$84956 - 43214 = 41742$$

Find **Q**:

$$41742 + 32405 = 74147$$

6. Correct Answer: A. 57

Aman's age: XXI = 21

- Raghav's age: $21 + 15 = 36$
- Sum of ages: $21 + 36 = 57$

7. Correct Answer: A. 45800

Round 95846 to nearest hundreds → **95800**

- Round 49876 to nearest thousands → **50000**
- Subtract:$95800 - 50000 = 45800$

8. Correct Answer: A. 11386

Total population = $4758 + 3978 + 2650$

$$4758 + 3978 = 8736$$
$$8736 + 2650 = 11386$$

9. Correct Answer: B. 11700

Round each to nearest hundred:

- 4375 → 4400
- 1872 → 1900
- 3488 → 3500
- 1937 → 1900

Add:$4400 + 1900 + 3500 + 1900 = 11700$

10. Correct Answer: A. 1756

Number of cages = $14048 \div 8$

$$14048 \div 8 = 1756$$

11. Correct Answer: A. 200

Total photos = $4 \times 10 \times 5$

$$4 \times 10 = 40$$
$$40 \times 5 = 200$$

12. Correct Answer: B. 8800

5462 → 5500 (nearest hundred)

3258 → 3300 (nearest hundred)

Add:5500 + 3300 = 8800

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Answer: (A)

Rounded: 43,726 ≈ 44,000 and 56,438 ≈ 56,000

Sum ≈ 1,00,000

14. Correct Answer: (A), (B), (C), (D)

Check each:

- 45,678 + 32,211 = 77,889
- 24,567 + 46,345 = 70,912
- 56,780 + 43,220 = 1,00,000
- 78,456 + 62,543 = 1,40,999

15. Correct Answer: (A), (B), (C)

(A) 348 ≈ 300

(B) 76 ≈ 80

(C) 348 × 76 ≈ 300 × 80 = 24,000

(D) 26,000

16. Correct Answer: (A), (B), (C)

(A) Break 25 = 20 + 5

(B) a × b = b × a (valid property)

(C) Multiply by 100, divide by 4 (since 25 = 100/4)

(D) Repeated subtraction

17. Correct Answer: (A) ₹144

8 apples = ₹96 ⇒ 1 apple = ₹12

12 apples = 12 × 12 = ₹144

ASSERTION AND REASON TYPE

18. Correct Answer: (A)

- Assertion (A): True
  (A 5-digit + 6-digit number gives minimum 110000 → 6 digits, maximum 1099998 → 7 digits)
- Reason (R): True
  (Sum can have at most one more digit than the larger number)
- Explanation:
  The reason correctly explains why the sum can be either 6 or 7 digits.

19. Correct Answer: (A)

- Assertion (A): True
  (Order does not change product → commutative property)
- Reason (R): True
  (7 × 3 = 3 × 7 is a correct example)
- Explanation:
  The reason correctly explains the assertion using an example.

20. Correct Answer: (D)

- Assertion (A): False
  (A 5-digit number ÷ 1-digit number will always give at least a 4-digit quotient.
  Example: 10000 ÷ 9 ≈ 1111 → 4-digit, not 2-digit)
- Reason (R): True
  (Dividing a larger number by a smaller number gives a smaller value than the original number)
- Explanation:
  The assertion is incorrect, but the reason is a true general statement.

21. Correct Answer: (D)

- Assertion (A): False
  (Unitary method cannot solve *all* types of story problems—only those based on direct proportion or equal distribution.)
- Reason (R): True
  (Unitary method involves finding the value of one unit first, then using it to find the required value.)
- Explanation:
  The reason is correct, but it does not justify the incorrect assertion.

MATRIX MATCHNG TYPE

22. Correct Answer: (a)–R, (b)–P, (c)–Q, (d)–S

Match Column-I with Column-II

(a) 7,245 ÷ 5 = 1,449 → (R)

(b) 2,345 × 3 = 7,035 → (P)

(c) 95,412 – 48,235 = 47,177 → (Q)

(d) 36,345 + 24,250 = 60,595 → (S)

23. Correct Answer: (a)–S, (b)–Q, (c)–R, (d)–P

Match Column-I with Column-II

(a) ₹50 ÷ 5 = ₹10 → (S)
(b) ₹64 ÷ 8 = ₹8 → (Q)
(c) ₹200 ÷ 10 = ₹20 → (R)
(d) ₹48 ÷ 4 = ₹12 → (P)

COMPREHENSION TYPE

Passage 1:

24. Correct Answer: (B)

Ravi estimated the total as ₹7,500.

25. Correct Answer: (B)

He changed ₹275 to a nearby number (₹300), which is rounding off.

26. Correct Answer: (A)

His estimate is higher because he rounded ₹275 up to ₹300, increasing the total.

Passage 2:

27. Correct Answer: (B)

Priya estimated: ₹500 × 125 = ₹62,500

28. Correct Answer: (B)

She estimated by rounding off the price of each t-shirt.

29. Correct Answer: (B)

Her estimate is higher because she rounded ₹450 up to ₹500.

3. Factors and multiples

Solutions

Worksheet – 1

1. **Correct Answer: B) A number that divides another number exactly.**

   Solution: A factor is any number that divides another number without leaving a remainder.

2. **Correct Answer: C) 5.**

   Solution: A prime number has exactly two distinct factors: 1 and itself. The only factors of 5 are 1 and 5.

3. **Correct Answer: C) 4.**

   Solution: A composite number has more than two factors. The factors of 4 are 1, 2, and 4.

4. **Correct Answer: A) The number itself.**

   Solution: The highest factor of any number is the number itself since it divides itself exactly.

5. **Correct Answer: B) 15 and 28.**

   Solution: The factors of 15 are 1, 3, 5, 15; the factors of 28 are 1, 2, 4, 7, 14, 28. They share no common factors other than 1.

6. **Correct Answer:**

   Solution: The prime factorization of 36 can be found by dividing by the smallest primes: $36 \div 2 = 18, 18 \div 2 = 9, and\ 9 \div 3 = 3$.

7. **Correct Answer: C) Perfect square.**

   Solution: A number has exactly three factors if it is a perfect square of a prime number (e.g., $1, p, p^2$ where p is prime).

8. **Correct Answer: B) 2 is a prime number.**

   Solution: 2 is the only even prime number, as it has exactly two factors: 1 and 2.

9. **Correct Answer: A) 1, 2, 3, 4, 6, 8, 12, 24.**

   Solution: The complete list of factors of 24 includes all numbers that divide 24 exactly.

10. **Correct Answer: B) Composite.**

    Solution: Multiplying a prime number by itself results in a composite number, as it has more than two factors.

Worksheet: 2

11. **Correct Answer: C) Factors that are prime numbers.**

    Solution: Prime factors are the factors of a number that are prime numbers themselves.

12. **Correct Answer: B) 7**

    Solution: A prime number has exactly two distinct factors: 1 and itself. 7 only has factors 1 and 7.

13. **Correct Answer: A) $2 \times 3 \times 5$.**

    Solution: The prime factorization of 30 is $2 \times 3 \times 5$, where 2,3, and 5 are prime.

14. **Correct Answer: C) 4.**

    Solution: The factors of 12 are 1, 2, 3, 4, 6, 12; the factors of 16 are 1, 2, 4, 8, 16.

15. **Correct Answer: B) 15 and 28.**

    Solution: The factors of 15 are 1, 3, 5, 15; the factors of 28 are 1, 2, 4, 7, 14, 28.

16. **Correct Answer: A) $2^3 \times 7$.**

    Solution:

    The prime factors for $56 = 2 \times 2 \times 2 \times 7$.

    Thus, $56 = 2^3 \times 7$.

17. **Correct Answer: A) 12.**

    Solution: $2^2 = 4$ and $3^1 = 3$.

    Therefore, $4 \times 3 = 12$.

Worksheet: 3

18. **Correct Answer: B) Numbers are obtained by multiplying the number by integers.**

    Solution: Multiples of a number are found by multiplying it by whole numbers (0, 1, 2, 3, ...).

19. **Correct Answer: B) 7.**

    Solution: The first multiple of any number is the number itself, so the first multiple of 7 is 7.

20. **Correct Answer: C) 30.**

    Solution: A number is a multiple of 5 if it ends in 0 or 5. Here, 30 ends in 0, so it is a multiple of 5.

21. **Correct Answer: C) 25.**

    Solution: To check for multiples of 3, the sum of the digits must be divisible by 3.

22. **Correct Answer: A) 12.**

    Solution: The multiples of 4 are 4, 8, 12, 16, ... and the multiples of 6 are 6, 12, 18, ... The smallest common multiple is 12

23. **Correct Answer: A) 12.**

    Solution: The least common multiple of 3 and 4 is 12, so any number that is a multiple of both must also be a multiple of 12.

24. **Correct Answer: B) 10.**

    Solution: The multiples of 10 from 10 to 100 are 10, 20, 30, ..., 100, which gives us 10 multiples.

Worksheet: 4

25. **Correct Answer: (B) 15**

    Solution: The multiples of 3 are 3,6,9,12,15,... and the multiples of 5 are 5,10,15,20,...

    The smallest common multiple is 15.

26. **Correct Answer: (B) 12**

    Solution: The multiples of 4 are 4,8,12,... and the multiples of 6 are 6,12,18,...

    The first common multiple is 12.

27. **Correct Answer: (D) A multiple of both numbers.**

    Solution: The LCM of two numbers is the smallest number that is divisible by both numbers.

28. **Correct Answer: (A) 24**

    Solution: The multiples of 6 are 6,12,18,24,..., and the multiples of 8 are 8,16,24,...

    The smallest common multiple is 24.

29. **Correct Answer: (C) 12**

    Solution: The multiples of 2, 3, and 4 are 12,24,36,...

    The smallest common multiple is 12.

30. **Correct Answer: (A) 30**

    Solution: The multiples of 5, 10, and 15 are 30,60,90,...

    The LCM is 30 because it is the smallest common multiple

31. **Correct Answer: (C) 12**

    Solution: Use the formula:

    $$LCM \times HCF = Product\ of\ the\ two\ numbers.$$

    Here, LCM=36, and HCF=9.

    $$LCM \times HCF = Product\ of\ the\ two\ numbers$$

    $$= 36 \times 9 = Product\ of\ the\ two\ numbers$$

    $$Product\ of\ the\ two\ numbers = 324.$$

    Therefore, the product of two numbers is 324.

Worksheet: 5

32. **Correct Answer: (C) The last digit is even.**

    Solution: For a number to be divisible by 2, its last digit should be one of these: 0, 2, 4, 6, or 8.

33. **Correct Answer: (B) 240**

    Solution: A number is divisible by 5 if its last digit is either 0 or 5. Among the given numbers, 240 ends in 0.

34. **Correct Answer: (A) A number must end with 0.**

    Solution: For a number to be divisible by 10, its last digit must be 0.

35. **Correct Answer: (C) divisible by 9**

    Solution: The rule for divisibility by 9 is that the sum of the digits must be divisible by 9.

    The sum of digits of 126 is $1 + 2 + 6 = 9$, which is divisible by 9.

36. **Correct Answer: (B) The sum of its digits is divisible by 3.**

Solution: The divisibility rule for 3 states that if the sum of a number's digits is divisible by 3, the number itself is divisible by 3.

37. **Correct Answer: (A) 1452**

Solution: For divisibility by 11, the difference between the sum of digits in odd places and even places must be divisible by 11.

For 1452, $(1 + 5) - (4 + 2) = 6 - 6 = 0$, which is divisible by 11.

38. **Correct Answer: (B) 6**

Solution: If a number is divisible by both 2 and 3, it must be divisible by their least common multiple, which is 6.

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: D**

Solution:
The factors of 24 are 1, 2, 3, 4, 6, 8, 12, 24.
Hence, 6, 8, and 12 are all factors of 24.

2. **Correct Answer: B**

Solution:
The smallest factor of any number is always

3. **Correct Answer: A. 100**

Given numbers: **5, 12, 8**

The **4th multiple** means multiply by 4.

- 4th multiple of 5 = $5 \times 4 = 20$
- 4th multiple of 12 = $12 \times 4 = 48$
- 4th multiple of 8 = $8 \times 4 = 32$

Sum = $20 + 48 + 32 = 100$

4. **Correct Answer: D. 9**

A number with exactly **3 factors** is usually a square of a prime number.

Check options:

- 4 → factors: 1, 2, 4 → **3 factors**
- 8 → factors: 1, 2, 4, 8 → 4 factors
- 5 → factors: 1, 5 → 2 factors
- 9 → factors: 1, 3, 9 → **3 factors**

Among 4 and 9, the greatest is **9**.

5. **Correct Answer: D**

Let us check each option:

D False → Example: difference of 5 and 2 = 3 (odd)

So this is the incorrect statement.

6. **Correct Answer: A) 100**

Sum of fourth multiples of the given numbers
The numbers shown are 5, 8, and 12.
Fourth multiples:

- $\mathbf{5 \times 4 = 20}$
- $\mathbf{8 \times 4 = 32}$
- $\mathbf{12 \times 4 = 48}$

Sum = $\mathbf{20 + 32 + 48 = 100}$

7. **Correct Answer: D) 9**

Greatest 1-digit number that has only 3 factors
A number with exactly 3 factors is a square of a prime number.

Options:

- 4 → factors: 1, 2, 4
- 8 → 1, 2, 4, 8
- 5 → 1, 5
- 9 → 1, 3, 9

Greatest = 9

8. **Correct Answer: D**

Select the INCORRECT option
Check option C:

6th multiple of 8 = $\mathbf{8 \times 6 = 48}$
9th multiple of 7 = $\mathbf{7 \times 9 = 63}$

Sum = $\mathbf{48 + 63 = 111}$

Check D:
Difference of two prime numbers is not always even
Example: $\mathbf{5 - 2 = 3}$(odd)

So D is incorrect.

9. **Correct Answer: A) 6**

Multiples of 15 between 95 and 185
Multiples: 105, 120, 135, 150, 165, 180

Total = 6

10. **Correct Answer: C (48, 6, 4)**

Fill in the blanks

- LCM of 24 and 48 = **48** $\rightarrow P = 48$
- Factors of 124 = 1,2,4,31,62,124$\rightarrow$ **6 factors** $\rightarrow Q = 6$
- HCF of 12 and 32 = **4** $\rightarrow R = 4$

11. **Correct Answer: C) 48**

Greatest number of factors

- 12 $\rightarrow$ 6 factors
- 24 $\rightarrow$ 8 factors
- 48 $\rightarrow$ 10 factors
- 32 $\rightarrow$ 6 factors

Greatest = **48**

12. **Correct Answer: D) 13**

Multiples of 5 from 20 to 80
20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80

Total = **13**

13. **Correct Answer: C) 34**

Sum of (3rd multiple of 11) and difference between common factors of 8 and 10
3rd multiple of 11 = $11 \times 3 = 33$

Common factors of 8 and 10 = **1, 2**
Difference = $2 - 1 = 1$

Sum = $33 + 1 = 34$

14. **Correct Answer: A**

Match the following

- **P:** Number of factors of 125
  $125 = 5^3$
  Number of factors = $3 + 1 = 4\rightarrow$ **(iv)**
- **Q:** Common prime factor of 15 and 50
  Factors:
  15 = 3 × 5
  50 = 2 × 5 × 5
  Common prime factor = **5** $\rightarrow$ **(i)**
- **R:** LCM of 16 and 34
  $16 = 2^4$, $34 = 2 \times 17$
  LCM = $2^4 \times 17 = 272\rightarrow$ **(ii)**
- **S:** HCF of 98 and 14
  = **14** $\rightarrow$ **(iii)**

So matching is:
P $\rightarrow$ (iv), Q $\rightarrow$ (i), R $\rightarrow$ (ii), S $\rightarrow$ (iii)

15. **Correct Answer: B) 6892**

Identify the number

Let ones digit = smallest prime = **2**

- Hundreds digit = 4th multiple of 2 = **8**
- Tens digit = 7 more than 2 = **9**
- Thousands digit = 3 less than 9 = **6**

Number = **6892**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. **Correct Answer: (A) 3, (B) 6, (C) 9**

Solution:
Factors of 36 are numbers that divide 36 completely.

$36 \div 3 = 12$,

$36 \div 6 = 6$,

$36 \div 9 = 4$.

Thus, 3, 6, and 9 are factors of 36, but 15 is not a factor.

17. **Correct Answer: (A) 9, (C) 15**

Solution:
Composite numbers have more than two factors.

- 9=1,3,9 (Composite)
- 13=1,13 (Prime)
- 15=1,3,5,15 (Composite)
- 23=1,23 (Prime)

18. **Correct Answer: (A) 8 and 9, (B) 15 and 16, (D) 7 and 13**

Solution:
Co-prime numbers have 1 as their only common factor.

- 8 and 9: Factors are 1,2,4,8 and 1,3,9; common factor is 1.
- 15 and 16: Factors are 1,3,5,15 and 1,2,4,8,16; common factor is 1.
- 12 and 18: Common factors are 1,2,3,6; not co-prime.

- 7 and 13: Both are prime numbers; only common factor is 1.

19. Correct Answer: (A) 18, (B) 24, (D) 36

Solution:
A number divisible by both 2 and 3 must be divisible by 6.

- 18÷6=3 (Yes)
- 24÷6=4 (Yes)
- 35÷6≠integer (No)
- 36÷6=6 (Yes)

20. Correct Answer: (A) 20, (B) 40, (C) 60, (D) 80

Solution:
Multiples of $4 \times 5 = 20$ are 20,40,60,80.
Thus, all options are correct as they are divisible by both 4 and 5.

ASSERTION AND REASON TYPE

21. Correct Answer: (C) Assertion (A) is true, and Reason (R) is false.

Explanation:

- Composite numbers have more than two factors. This makes the assertion true.
- However, the reason is false because composite numbers are divisible by more than just 1 and themselves; they are also divisible by other factors.

16. Correct Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

Explanation:

- The number 1 is neither prime nor composite because it does not meet the definition of either type.
- A prime number must have exactly two factors, and 1 only has one factor (itself). The reason explains this correctly.

22. Correct Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

Explanation:

- The HCF of two numbers divides their LCM perfectly, which makes the assertion true.
- Additionally, the relationship $HCF \times LCM = Product\ of\ the\ two\ numbers$ ensures that the HCF is always a factor of the LCM.

MATRIX MATCHNG TYPE

23. Correct Answer: (a) → (R), (b) → (P), (c) → (S), (d) → (Q)

Explanation:

1. Factors of 24 are 1,2,3,4,6,8,12,24.
2. Prime numbers less than 10 are 2,3,5,7.
3. The smallest composite number is 4, as it has factors other than 1 and itself.
4. The smallest prime number is 2.

24. Correct Answer: (a) → (R), (b) → (Q), (c) → (P), (d) → (S)

Explanation:

1. The smallest multiple of 6 is 6.
2. Multiples of 4 are 4,8,12,16,20,...
3. Common multiples of 4 and 6 are 12,24,36,...
4. The smallest common multiple (LCM) of 4 and 6 is 12.

COMPREHENSION TYPE

Passage-1

25. Correct Answer: B) HCF

That mathematical concept is called HCF (Highest Common Factor).

- Factors of 36 = 1, 2, 3, 4, 6, 9, 12, 18, 36
- Factors of 48 = 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

The highest common factor is 12.

So, Ravi should use HCF to find the largest equal number of kits/items possible.

26. Correct Answer: C) 12

Solution:
Factors of 36 = 1, 2, 3, 4, 6, 9, 12, 18, 36
Factors of 48 = 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

Highest common factor = **12**

27. Correct Answer: D) 12

solution:
Using HCF = **12**
Number of kits = **12**

Passage-2

28. Correct Answer: B) Check if the sum of digits is divisible by 3

Explanation:
For divisibility by 3, add the digits:
2 + 4 + 0 = 6
Since **6 is divisible by 3**, 240 is divisible by 3.

29. Correct Answer: A) Yes

Explanation:
A number is divisible by 5 if it ends in **0 or 5**.
240 ends with **0**.

30. Correct Answer: C) 20

Solution:
240 ÷ 12 = 20

4. Fractions and Decimals

Solutions

Worksheet – 1

1. **Correct Answer: (C) $\frac{1}{3}$**

   Total eggs = 9
   Bad eggs = 6 ⟶ Good eggs = 9 − 6 = 3
   Fraction of good eggs = $\frac{3}{9} = \frac{1}{3}$

2. **Correct Answer: (B) $\frac{1}{4}$**

   Proper fraction means numerator < denominator
   Only $\frac{1}{4}$ satisfies this

3. **Correct Answer: (A) $\frac{2}{4}$**

   Equivalent fraction of $\frac{1}{2}$: multiply numerator & denominator by same number

   $$\frac{1 \times 2}{2 \times 2} = \frac{2}{4}$$

4. **Correct Answer: (C) $\frac{7}{12}$**

   Total parts = 12
   Green shaded parts = 7
   Fraction = $\frac{7}{12}$

5. **Correct Answer: (A) $3\frac{4}{5}$**

   Convert $\frac{19}{5}$ to mixed fraction
   19 ÷ 5 = 3 remainder 4
   So, $3\frac{4}{5}$

6. **Correct Answer: (A) $\frac{11}{3}$**

   Convert $3\frac{2}{3}$ to improper fraction
   $(3 \times 3) + 2 = 11 \rightarrow \frac{11}{3}$

7. **Correct Answer: (B) $\frac{7}{11}$**

   Simplest form = cannot be reduced further

   - $\frac{12}{14} = \frac{6}{7}$ (can reduce)
   - $\frac{7}{11}$ (already simplest)
   - $\frac{4}{6} = \frac{2}{3}$ (can reduce)
   - $\frac{4}{8} = \frac{1}{2}$ (can reduce)

Worksheet – 2

1. **Correct Answer: (A) $\frac{5}{6}$**

   Rahul ate $\frac{2}{6}$, Anu ate $\frac{3}{6}$
   Total eaten = $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$
   Correct Answer: (A) $\frac{5}{6}$

2. **Correct Answer: (D) 9**

   $\frac{7}{12} + \frac{2}{12} = \frac{9}{12}$
   Missing number = 9

3. **Correct Answer: (C) $\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$**

   First circle = $\frac{3}{4}$, second = $\frac{2}{4}$
   Subtract: $\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$

4. **Correct Answer: (B) $\frac{5}{6}$**

   Ravi walked $\frac{1}{2}$km and $\frac{1}{3}$km
   Find LCM of 2 and 3 = 6

   $$\frac{1}{2} = \frac{3}{6}, \frac{1}{3} = \frac{2}{6}$$

   Total distance = $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

5. **Correct Answer: (B) $\frac{1}{2}$**

   Juice = $\frac{5}{6}$, Sita drinks $\frac{1}{3} = \frac{2}{6}$
   Left = $\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$

6. **Correct Answer: (A) $\frac{1}{3}$**

   Tank is $\frac{2}{3}$ full, used = $\frac{1}{2}$ of it
   Used = $\frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$

7. **Correct Answer: (A) $\frac{1}{6}$**

   Total chocolate = $\frac{5}{6}$, shared among 5 children
   Each gets = $\frac{5}{6} \div 5 = \frac{5}{6} \times \frac{1}{5} = \frac{5}{30} = \frac{1}{6}$

Worksheet – 3

1. **Correct Answer: (B) Aman**
   Same denominator (7): compare numerators
   5 > 3 ⇒ Aman drank more
2. **Correct Answer: (A) Riya**
   Compare $\frac{5}{8}$ and $\frac{3}{5}$
   Cross multiply:
   5×5 = 25, 3×8 = 24 ⇒ 25 > 24
   So $\frac{5}{8} > \frac{3}{5}$
3. **Correct Answer: (B) 4/5**
   Compare $\frac{7}{9}$ and $\frac{4}{5}$
   Cross multiply:
   7×5 = 35, 4×9 = 36 ⇒ 35 < 36
   So $\frac{4}{5}$ is greater
4. **Correct Answer: (A) 4/5**
   Compare with $\frac{5}{6}$:
   $$\frac{4}{5} = 0.8 < 0.83$$
   Others are greater than $\frac{5}{6}$
5. **Correct Answer: (B) Bottle B**
   Compare $\frac{1}{2}$ and $\frac{3}{4}$
   $$\frac{3}{4} > \frac{1}{2}$$
6. **Correct Answer: (B) 1/6, 3/6, 5/6**
   Same denominator (6): compare numerators
   1 < 3 < 5 ⇒ ascending order
7. **Correct Answer: (B) 6/7, 4/7, 2/7**
   Same denominator (7): compare numerators
   6 > 4 > 2 ⇒ descending order

Worksheet – 4

1. **Correct Answer: (A) 1/4**
   What is 0.25 written as a fraction?
   0.25 = 25/100 = 1/4
2. **Correct Answer: (A) 1.3 and 2.6**
   Example of like decimals
   Like decimals have same number of decimal places.
   1.3 and 2.6 → both have 1 decimal place
3. **Correct Answer: (C) 0.875**
   Decimal for 7/8
   7 ÷ 8 = 0.875
4. **Correct Answer: (A) 0.6**
   Compare 0.6 and 0.56
   0.6 = 0.60 → 0.60 > 0.56
5. **Correct Answer: (A) 3/8**
   Express 0.375 as a fraction
   0.375 = 375/1000 = 3/8
6. **Correct Answer: (A) 0.005**
   Convert 5/1000 into decimal
   5 ÷ 1000 = 0.005
7. **Correct Answer: (B) 72/100 = 0.72**
   72 out of 100 in decimal
   72/100 = 0.72

Worksheet – 5

1. **Correct Answer: (A) 20.77**
   What is 12.3 + 8.47?
   12.30 + 8.47 = 20.77
2. **Correct Answer: (B) 10.43 km**
   Total distance covered
   5.65 + 4.78 = 10.43 km
3. **Correct Answer: (A) 4.85 litres**
   Water left
   8.6 − 3.75 = 4.85 litres
4. **Correct Answer: (A) 65.25**
   Find 35.87 + 49.63 − 20.25
   35.87 + 49.63 = 85.50
   85.50 − 20.25 = 65.25
5. **Correct Answer: (D) 11.75 kg**
   Rice left
   25.4 + 13.7 = 39.1
   50.85 − 39.1 = 11.75 kg
6. **Correct Answer: (A) 7.28 meters**
   Fabric remaining
   15.3 + 9.87 = 25.17
   32.45 − 25.17 = 7.28 meters
7. **Correct Answer: (B) 99.6**
   Calculate 125.87 − 68.9 + 42.63
   125.87 − 68.9 = 56.97
   56.97 + 42.63 = 99.6

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (C) 4**
   More parts to shade
   Already shaded = $\frac{2}{10} = \frac{1}{5}$
   Needed = $\frac{3}{5}$
   Extra = $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$

Total parts = 10
$\frac{2}{5}$of 10 = 4 parts

2. **Correct Answer: (A) 1/5**
Ribbon in each piece
$$\frac{4}{5} \div 4 = \frac{4}{5} \times \frac{1}{4} = \frac{4}{20} = \frac{1}{5}$$
3. **Correct Answer: (A) 0.6**
Water left
0.9 – 0.3 = 0.6 L
4. **Correct Answer: (B) 5.75**
Total distance walked
3.4 + 2.35 = 5.75 km
5. **Correct Answer: $\frac{1}{5}$(Option D)**
Total spent = $\frac{5}{10} + \frac{3}{10} = \frac{8}{10}$
Saved = $1 - \frac{8}{10} = \frac{2}{10} = \frac{1}{5}$
6. **Correct Answer: C**
Ascending order means smallest → largest.
   - A: Not in order (5/12 comes after 13/12)
   - B: Not in order (12/9 should come earlier)
   - C: $4/18 < 5/18 < 6/18 < 12/18 < 19/18$
   - D: Not in order (10/24 should come before 12/24)
7. **Correct Answer: D. None of these**
Figure P: 6 out of 12 parts shaded → $\frac{6}{12} = \frac{1}{2}$
   - Figure Q: 3 out of 6 parts shaded → $\frac{3}{6} = \frac{1}{2}$
   - Figure R: 2 out of 5 parts shaded → $\frac{2}{5}$

Now check options:
   - A: $\frac{1}{2} + \frac{2}{5} = \frac{5}{10} + \frac{4}{10} = \frac{9}{10}$
   - B: Unshaded P = $\frac{1}{2}$, Shaded Q = $\frac{1}{2}$
   - C: Q = $\frac{1}{2}$, R = $\frac{2}{5}$, so $\frac{1}{2} > \frac{2}{5}$

All A, B, C are correct.
8. **Correct Answer: 200 (Option C)**
Males = $\frac{3}{7} \times 350 = 150$
Females = $350 - 150 = 200$
9. **Correct Answer: $\frac{1}{4}$(Option B)**
Total sold = $40 + 80 = 120$
Fraction sold = $\frac{120}{480} = \frac{1}{4}$
10. **Correct Answer: 159 (Option C)**
Unread part = $1 - \frac{8}{9} = \frac{1}{9}$
Pages left = $\frac{1}{9} \times 1431 = 159$
11. **Correct Answer: C**
From the figure:
   - Total parts = 8
   - Red parts = 3 → $\frac{3}{8}$
   - Blue parts = 2
   - Green parts = 3

   So, ratio = $\frac{2}{3}$
12. **Correct Answer: $\frac{8}{15}$(Option C)**
Days in Shimla = $15 - 7 = 8$
Fraction = $\frac{8}{15}$
13. **Correct Answer: $\frac{7}{10}$(Option C)**
Stickers left = $500 - 150 = 350$
Fraction left = $\frac{350}{500} = \frac{7}{10}$
14. **Correct Answer: 5 (Option A)**
One whole = $\frac{5}{5}$
Each part = $\frac{1}{5}$
Number of parts = 5

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answers: (A), (C), (D)**
Correct calculations
   - $\frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$
   - $\frac{2}{5} - \frac{1}{5} = \frac{1}{5} \neq \frac{5}{5}$
   - $\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$
   - $\frac{5}{6} - \frac{1}{2} = \frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$
16. **Correct Answers: (A), (C)**
Fractions greater than $\frac{1}{2}$
   - $\frac{3}{4} = 0.75$
   - $\frac{2}{5} = 0.4$
   - $\frac{5}{4} = 1.25$
   - $\frac{1}{5} = 0.2$
17. **Correct Answers: (A), (B), (D)**
Valid conversions
   - $\frac{1}{2} = 0.5$
   - $\frac{3}{4} = 0.75$

- $\frac{1}{4} = 0.25 \neq 1.33$
- $\frac{5}{10} = 0.5$

18. **Correct Answers: (A), (B), (C)**
Correct calculations
    - 1.5 + 2.5 = 4.0
    - 3.6 – 1.2 = 2.4
    - 0.75 + 0.25 = 1.0
    - 2.8 – 0.5 = 2.3 ≠ 1.3
19. **Correct Answers: (A), (B), (D)**
Decimals between 0 and 1
    - 0.2
    - 0.75
    - (greater than 1)
    - 0.5

ASSERTION AND REASON TYPE

20. **Correct Answer: (A)**
Assertion–Reason
Assertion: $\frac{5}{8} = 0.625$, $\frac{3}{4} = 0.75 \rightarrow 0.625 < 0.75$ (True)
Reason: Converting fractions to decimals helps compare (True)
Reason correctly explains the assertion
21. **Correct Answer: (D)**
Assertion–Reason
Assertion: 0.3 = 0.30 and 0.30 > 0.25 (False)
Reason: Decimals are compared digit by digit (True)
22. **Correct Answer: (D) (Assertion is false, Reason is true)**
Assertion–Reason
Assertion (A):
$$\frac{3}{4} - \frac{1}{3}$$
LCM of 4 and 3 = 12
$= \frac{9}{12} - \frac{4}{12} = \frac{5}{12} \neq \frac{1}{12}$ (False)
Reason (R):
Subtraction of unlike fractions requires a common denominator (LCM) (True)

MATRIX MATCHING TYPE

23. **Correct Answer: (a)–Q, (b)–R, (c)–S, (d)–P**
Match Column-I with Column-II
    - Proper Fraction → numerator < denominator → $\frac{13}{17}$ → (a)–(Q)
    - Improper Fraction → numerator ≥ denominator → $\frac{17}{14}$ → (b)–(R)
    - Equivalent Fractions → equal value → $\frac{9}{18} = \frac{1}{2}$ → (c)–(S)
    - Whole Number → $\frac{19}{19} = 1$ → (d)–(P)
24. **Correct Answer: (a)–Q, (b)–R, (c)–S, (d)–P**
Match Column-I with Column-II
    - 2.4 + 3.35 = 5.75 → (a)–(Q)
    - 7 – 2.5 = 4.5 → (b)–(R)
    - 6.8 + 1.2 = 8.0 → (c)–(S)
    - 4 – 0.75 = 3.25 → (d)–(P)

COMPREHENSION TYPE

Passage 1:

25. **Correct Answer: (B) 3/8**

Fraction left after Maya and Rohan ate
Maya = $\frac{3}{8}$, Rohan = $\frac{1}{4} = \frac{2}{8}$
Total eaten = $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$
Left = $1 - \frac{5}{8} = \frac{3}{8}$

26. **Correct Answer: (B) 0.25**
Decimal of $\frac{1}{4}$
$$\frac{1}{4} = 0.25$$
27. **Correct Answer: (B) 0.625**
Total eaten (in decimal)
$$\frac{5}{8} = 0.625$$

Passage 2:

28. **Correct Answer: (A) 10.25 km**
Total distance run
3.5 + 2.75 + 4 = 10.25 km
29. **Correct Answer: (B) Yes, by 0.25 km**
Did Arjun run more than 10 km?
10.25 – 10 = 0.25 km → Yes
30. **Correct Answer: (B) 11/4**
Convert 2.75 into a fraction
2.75 = 275/100 = 11/4

5. Time and Money

Solutions

Worksheet – 1

1. **Correct Answer:**

   Hour hand on 4 and minute hand on 12 → Time = 4:00
   Correct Answer: A: 4:00

2. **Correct Answer: B: 4:00**

   3:15 + 45 minutes = 4:00

3. **Correct Answer: A: 30 minutes**

   2:00 to 2:30 = 30 minutes

4. **Correct Answer: B: 80 minutes**

   1 hour 20 minutes = 60 + 20 = 80 minutes

5. **Correct Answer: B: 7 hours 35 mins**

   From 10:30 a.m to 6:05 p.m
   → 10:30 to 5:30 = 7 hours
   → 5:30 to 6:05 = 35 minutes
   → Total = 7 hours 35 minutes

6. **Correct Answer: A: 4500 seconds**

   1 hour 15 minutes in seconds
   → 1 hour = 3600 seconds
   → 15 minutes = 15 × 60 = 900 seconds
   → Total = 3600 + 900 = 4500 seconds

7. **Correct Answer: C: 45 minutes**

   From 8:45 AM to 9:30 AM
   → 8:45 to 9:00 = 15 minutes
   → 9:00 to 9:30 = 30 minutes
   → Total = 45 minutes

Worksheet – 2

1. **Correct Answer: A) 18:00**

   6:00 PM → Add 12 hours to convert into 24-hour format
   6 + 12 = 18

   So, time = 18:00

2. **Correct Answer: C) 12:00**

   12:00 noon in 24-hour format = 12:00

3. **Correct Answer: A) 15:00**

   03:00 PM → 3 + 12 = 15:00

4. **Correct Answer: A) 2:45 PM**

   14:45 → 14 – 12 = 2:45 PM

5. **Correct Answer: B) 11:15 PM**

   23:15 → 23 – 12 = 11:15 PM

6. **Correct Answer: A) 21:00**

   7:00 PM → 19:00; add 2 hours → 21:00

7. **Correct Answer: A) 06:30**

   06:30 (morning, AM) → 06:30

Worksheet – 3

1. **Correct Answer: B) 2 hours 30 minutes**

   9:00 AM to 11:30 AM
   = 2 hours 30 minutes

2. **Correct Answer: A) 1:30 PM**

   3:45 PM – 2 hours 15 minutes
   = 1:30 PM

3. **Correct Answer: A) 2 hours and 35 minutes**

   2:40 PM to 5:15 PM
   = 2 hours 35 minutes

4. **Correct Answer: A) 1 hour 45 minutes**

   10:15 AM to 12:00 noon
   = 1 hour 45 minutes

5. **Correct Answer: D) 9:15 AM**

   7:25 AM + 1 hour = 8:25 AM
   8:25 AM + 50 minutes = 9:15 AM

6. **Correct Answer: A) 1:30 PM**

   4:10 PM – 2 hours = 2:10 PM
   2:10 PM – 40 minutes = 1:30 PM

7. **Correct Answer: A) 3:30 PM**

   11:00 AM + 4 hours = 3:00 PM
   3:00 PM + 30 minutes = 3:30 PM

Worksheet – 4

1. **Correct Answer: A) 575 paise**

   ₹5.75 = 575 paise

2. **Correct Answer: B) 825 paise**

   ₹8.25 = 825 paise

3. **Correct Answer: A) ₹12**

   1200 paise = ₹12

4. **Correct Answer: B) 750 paise**

   ₹7.50 = 750 paise

5. **Correct Answer: A) ₹5.00**

   ₹3.20 = 320 paise; 320 + 180 = 500 paise = ₹5.00

6. **Correct Answer: D) ₽0.00**

   1250 paise = ₹12.50; ₹12.50 − ₹12.50 = ₹0.00

7. **Correct Answer: A) ₽56.25**

   ₹10.75 + ₹45.50 = ₹56.25

Worksheet – 5

8. **Correct Answer: B) ₹28.25**

   ₹15.75 + ₹12.50 = ₹28.25

9. **Correct Answer: B) ₹16.25**

   ₹25.00 − ₹8.75 = ₹16.25

10. **Correct Answer: A) ₹22.50**

    3 × ₽7.50 = ₹22.50

11. **Correct Answer: A) ₹8.10**

    ₹32.40 ÷ 4 = ₹8.10

12. **Correct Answer: C) ₹48.25**

    ₹35.50 + ₹12.75 = ₹48.25

13. **Correct Answer: A) ₹24.10**

    ₹120.50 ÷ 5 = ₽24.10

14. **Correct Answer: A) ₹8.00**

    ₽24.00 ÷ 3 = ₽8.00

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C) 9:10 AM**

   Total time = 35 + 45 + 10 = 90 minutes = 1 hour 30 minutes
   7:40 AM + 1 hour 30 minutes = 9:10 AM

2. **Correct Answer: B) ₽70**

   ₽1000 ÷ 4 = ₽250 each
   ₽250 − ₽180 = ₽70

3. **Correct Answer: A) ₹1,200**

   Total money = ₽5,000
   Administrative fee = ₽200

   Remaining = 5000 − 200 = ₽4,800
   Each sibling = 4800 ÷ 4 = ₽1,200

4. **Correct Answer B: 2 hours 25 minutes**

   Start time = 5:35 p.m.
   End time = 8:00 p.m.

   From 5:35 to 6:00 = 25 minutes
   From 6:00 to 8:00 = 2 hours

   Total duration = 2 hours 25 minutes

5. **Correct Answer: A. ₹5725**

   Cost of 20 chairs = 20 × 110 = ₽2200
   Cost of 15 tables = 15 × 235 = ₽3525

   Total cost = 2200 + 3525 = ₽5725

6. **Correct Answer A. ₹660**

   Cost of 1 sack = ₹120 (from image)

   Cost of 18 sacks = 18 × 120 = ₹2160

   Money she has = ₹1500

   Extra money needed = 2160 − 1500 = ₹660

7. **Correct Answer D. 3:00 p.m**

   Start time = 1:15 p.m.

   Add 1 hour → 2:15 p.m.
   Add 45 minutes → 3:00 p.m.

   Final time = 3:00 p.m.

8. **Correct Answer: A. ₹77230**

   Clothes = ₹6420
   Jewellery = ₹67940
   Footwear = ₹2870

   Total = 6420 + 67940 + 2870
   = 6420 + 2870 = 9290
   = 9290 + 67940 = ₹77230

9. **Correct Answer: B. 135 minutes**

   Time for 1 picture = $\frac{1}{4}$hour = 15 minutes

   Time for 9 pictures = 9 × 15 = 135 minutes

10. **Correct Answer: B. 5:45 p.m.**

    Bus time = 6:25 p.m.

    Subtract 40 minutes:
    6:25 – 25 minutes = 6:00 p.m.
    6:00 – 15 minutes = 5:45 p.m.

11. **Correct Answer: C. ₹105**

    Cost of 5 roses = 5 × 15 = ₹75
    Cost of 3 sunflowers = 3 × 10 = ₹30

    Total cost = 75 + 30 = ₹105

12. **Correct Answer: A. 20 mins past 8**

    Clock shows 7:50 (minute hand at 10 = 50 minutes, hour hand near 7).

    Actual time is 30 minutes more:
    7:50 + 30 minutes = 8:20

    8:20 = 20 minutes past 8

13. **Correct Answer: D. ₽100**

    2 drinks = ₹130 ⇒ 1 drink = ₹65

    Now,2 popcorn + 1 drink = ₹265
    2 popcorn + 65 = 265
    2 popcorn = 200
    1 popcorn = ₽100

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

14. **Correct Answer: A, B, C**
    - 12:00 AM = midnight
    - 12:00 PM = noon
    - 3:00 PM = afternoon
    - 12:00 PM ≠ midnight

15. **Correct Answer: A, B, C, D**
    - 15:00 = 3:00 PM
    - 11:00 = 11:00 AM
    - 19:00 = 7:00 PM
    - 12:00 = 12:00 PM (noon)

16. **Correct Answer: A, B, C**
    - 6:00 PM → 8:00 PM = 2 hours
    - 5:30 AM → 7:15 AM = 1 hour 45 minutes
    - 9:00 AM → 9:50 AM = 50 minutes
    - 10:00 PM → 9:30 PM is not possible (time cannot be negative)

17. **Correct Answer: A, B**
    - ₹1 = 100 paise
    - ₹5 = 500 paise
    - ₹100 = 10 paise
    - ₹50 = 500 paise

18. **Correct Answer: A, D**
    - 1000 paise = ₹10
    - 100 paise = ₹1
    - 1000 cents (not Indian currency)
    - 10 rupees = ₹10

19. **Correct Answer: A, C**

    Total amount = ₹1200, number of people = 6
    Each share = 1200 ÷ 6 = ₹200

    A. Each gets ₹200
    B. Each gets ₹150
    C. Total amount remains ₹1200
    D. Each gets ₹250

ASSERTION AND REASON TYPE

20. **Correct Answer: A**

    Assertion (A): True (Money can be converted into smaller units)
    - Reason (R): True (₹1 = 100 paise)
    - Reason correctly explains the assertion

21. **Correct Answer: B**

    Assertion (A): True (Travel time is measured in hours and minutes)
    - Reason (R): True (Elapsed time means total duration between start and end)
    - But Reason does not directly explain Assertion

22. **Correct Answer: A**

    Assertion (A): True (6:00 PM to 8:00 PM)
    - Reason (R): True (Time difference = 2 hours)
    - Reason correctly explains the assertion

MATRIX MATCHING TYPE

23. **Correct Answer: (a–P), (b–Q), (c–S), (d–R)**
    - 3 hours 30 minutes = $3 \times 60 + 30 =$ 210minutes → (P)
    - 1 hour 45 minutes = $60 + 45 =$ 105minutes → (Q)
    - 120 minutes = $120 \div 60 =$ 2hours → (S)
    - 300 minutes = $300 \div 60 =$ 5hours → (R)
24. **Correct Answer: (a–P), (b–R), (c–Q), (d–S)**
    - ₹500 + ₹250 = ₹750 → (P)
    - ₹800 − ₹300 = ₹500 → (R)
    - ₹120 × 5 = ₹600 → (Q)
    - ₹900 ÷ 3 = ₹300 → (S)

COMPREHENSION TYPE

Passage 1:

25. **Correct Answer: B 1 hour 45 minutes**

    First session: 9:30 AM → 11:15 AM
    = 1 hour 45 minutes
26. **Correct Answer: C 2 hours 15 minutes**

    Second session: 11:45 AM → 2:00 PM
    = 2 hours 15 minutes
27. **Correct Answer: B 4 hours**

    Total time = First session + Second session
    = 1 hour 45 minutes + 2 hours 15 minutes
    = 4 hours

Passage 2:

28. **Correct Answer: C ₹200**

    Total spent = ₹45 + ₹150 + ₹5 = ₹200
29. **Correct Answer: B ₹300**

    Money given = ₹500
    Spent = ₹200
    Change = 500 − 200 = ₹300
30. **Correct Answer: D 20000 paise**

    ₹200 = 200 × 100 = 20000 paise

6. Time and Money

Solutions

Worksheet – 1

1. **Correct Answer: (B) Millimeter**

   Explanation: The millimeter is the smallest unit of length in the metric system.

2. **Correct Answer: (A) 20 mm**

   Explanation: 1 centimeter = 10 millimeters.

   So, 2 centimeters = 2 × 10 = 20 millimeters.

3. **Correct Answer: (A) 0.5 kilometers**
   Explanation: 1 kilometer = 1000 meters.

   Therefore, 500 meters = 500 ÷ 1000 = 0.5 kilometers.

4. **Correct Answer: (C) Meter**

   Explanation: The meter is the most appropriate unit to measure the length of a classroom.

5. **Correct Answer: (A) 1.5 meters**

   Explanation: 1 meter = 100 centimeters. So, 150 centimeters = 150 ÷ 100 = 1.5 meters.

6. **Correct Answer: A) 1, 2, 3, 4, 6, 8, 12, 24.**

   Solution: The complete list of factors of 24 includes all numbers that divide 24 exactly.

7. **Correct Answer: (B) 5000 meters**

   Explanation: 1 kilometer = 1000 meters.

   Therefore, 5 kilometers = 5 × 1000 = 5000 meters.

Worksheet-2

1. **Correct Answer: (A) Gram**

   Explanation: Weight is measured in **grams (g)** in the metric system.

2. **Correct Answer: (A) 1,000 grams**

   Explanation: 1 kilogram = 1,000 grams.

3. **Correct Answer: (D) 3,000 grams**

   Explanation: 1 kilogram = 1,000 grams, so 3 kilograms = 3 × 1,000 = 3,000 grams.

4. **Correct Answer: (B) 0.15 kg**

   Explanation: 150 grams = 150 ÷ 1,000 = 0.15 kilograms.

5. **Correct Answer: (C) 4 kg**

   Explanation: Total weight = 2.5 kg + 1.5 kg = 4 kg.

6. **Correct Answer: (B) 1.2 kg**

   Explanation: 1,200 grams = 1,200 ÷ 1,000 = 1.2 kilograms.

7. **Correct Answer: (A) 0.5 kg**

   Explanation: 500 grams = 500 ÷ 1,000 = 0.5 kilograms.

Worksheet-3

1. **Correct Answer: (B) Liter**
   Explanation: The unit used to measure the volume of a liquid is **liter (L)**.
2. **Correct Answer: (C) 2,500 ml**
   Explanation: 1 liter = 1,000 milliliters, so 2.5 liters = 2.5 × 1,000 = 2,500 milliliters.
3. **Correct Answer: (A) 3 L**
   Explanation: 1 liter = 1,000 milliliters, so 3,000 milliliters = 3,000 ÷ 1,000 = 3 liters.
4. **Correct Answer: (C) 0.15 liters**
   Explanation: 150 milliliters = 150 ÷ 1,000 = 0.15 liters.
5. **Correct Answer: C) 135.**
   Solution: The first five multiples of 9 are 9, 18, 27, 36, and 45.
   Their sum is 9+18+27+36+45=135.
6. **Correct Answer: (A) 64 cm^3**
   Explanation: The volume of a cube = $side^3$ = 4 × 4 × 4 = 64 cm^3.
7. **Correct Answer: (A) 30 cm^3**
   Explanation: The volume of a rectangular box = length × width × height = 5 × 3 × 2 = 30 cm^3.

Worksheet-4

1. **Correct Answer: (A) 16 cm²**
   Explanation: The area of a square = side × side.
   So, area = 4 × 4 = 16 cm².
2. **Correct Answer: (B) 35 cm²**
   Explanation: The area of a rectangle = length × width.
   So, area = 7 × 5 = 35 cm².
3. **Correct Answer: (A) 9 cm**
   Explanation: The area of a square = side × side.
   So, side length = √81 = 9 cm.
4. **Correct Answer: (A) 60 cm²**
   Explanation: The area of a rectangle = length × width.
   So, area = 10 × 6 = 60 cm².
5. **Correct Answer: (A) 12 cm**
   Explanation: The area of a square = side × side.
   So, side length = √144 = 12 cm
6. **Correct Answer: (C) 8 cm**
   Explanation: The area of a rectangle = length × width.
   So, width = 72 ÷ 9 = 8 cm.
7. **Correct Answer: (A) 60 m²**
   Explanation: The area of a rectangle = length × width.
   So, area = 12 × 5 = 60 m².

Worksheet-5

1. **Correct Answer: (B) 20 cm**
   Explanation: The perimeter of a square = 4 × side length.
   So, perimeter = 4 × 5 = 20 cm.
2. **Correct Answer: (B) 22 cm**
   Explanation: The perimeter of a rectangle = 2 × (length + width).
   So, perimeter = 2 × (8 + 3) = 2 × 11 = 22 cm.
3. **Correct Answer: (B) 9 cm**
   Explanation: The perimeter of a square = 4 × side length.
   Therefore, side length = 36 ÷ 4 = 9 cm.
4. **Correct Answer: (A) 5 cm**
   Explanation: The perimeter of a rectangle = 2 × (length + width).
   30 = 2 × (10 + width).
   30 = 20 + 2 × width.
   Subtract 20 from both sides:
   10 = 2 × width.
   Divide both sides by 2:
   width = 5 cm.
5. **Correct Answer: (B) 24 cm**
   Explanation: The perimeter of a polygon = number of sides × length of each side.
   So, perimeter = 6 × 4 = 24 cm.
6. **Correct Answer: (A) 38 cm**
   Explanation: The perimeter of a rectangle = 2 × (length + width).
   So, perimeter = 2 × (12 + 7) = 2 × 19 = 38 cm.
7. **Correct Answer: (A) 200 cm**
   Explanation: First, find the perimeter of the rectangle:
   Perimeter = 2 × (6 + 4) = 2 × 10 = 20 meters.
   1 meter = 100 cm, so 20 meters = 20 × 100 = 200 cm.

COMPETETIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: A**
   Solution:
   1 meter is equal to 100 centimeters.
2. **Correct Answer: B**
   Solution:
   The area of a rectangle is calculated as:
   Area = Length × Width = 5 cm × 3 cm = 15 cm²
3. **Correct Answer: A**
   Solution:
   2 kilogram is equal to 2000 grams.
4. **Correct Answer: B**
   Solution:
   1 liter = 1000 milliliters, so 2 liters = 2 × 1000 = 2000 milliliters.
5. **Correct Answer: A**
   Solution:
   The perimeter of a square is calculated as:
   Perimeter = 4 × Side = 4 × 6 cm = 24 cm
6. **Correct Answer: A. >**
   Compare the weights
   15 kg 105 g + 7 kg 900 g
   = 23 kg 5 g

$$18 \text{ kg } 700 \text{ g} + 950 \text{ g}$$
$$= 19 \text{ kg } 650 \text{ g}$$

Since 23 kg 5 g > 19 kg 650 g

7. **Correct Answer: A. 250 g**
   Weight of each cube
   Total weight on the other side:
   $$500 + 500 + 500 = 1500 \text{ g}$$
   There are **6 cubes**.
   $$1500 \div 6 = 250 \text{ g}$$
8. **Correct Answer: C. 900 g**
   Find the weight of
   From the three scales:
   - □ + 🧸 + 🚗 = **2350 g**
   - □ + 🧸 = **1500 g**
   - 🚗 + 🧸 = **1750 g**

   Add the last two:
   (rock + bear) + (car + bear) = 3250
   Now subtract the first:
   3250 - 2350 = 900
   So,
   bear = 900 g
9. **Correct Answer: C. 420 mL**
   Juice in each glass
   Total juice:5 L 40 mL = 5040 mL
   Divide into 12 glasses: $5040 \div 12 = 420$
10. **Correct Answer: D. 1 L 200 mL**
    Juice left in the jug
    Total juice: $8 \times 250 = 2000$ mL = 2 L
    Consumed: 800 mL
    Left: $2000 - 800 = 1200$ mL
    $$1200 \text{ mL} = 1 \text{ L } 200 \text{ mL}$$
11. **Correct Answer: B. 8**
    Number of glasses
    Total water: 3 L 200 mL = 3200 mL
    Each glass: 400 mL
    $$3200 \div 400 = 8$$
12. **Correct Answer: D. 122 L 150 mL**
    Water filled in 7 days
    Per day:17 L 450 mL
    For 7 days: $17 \times 7 = 119$ L
    $450 \times 7 = 3150$ mL = 3 L 150 mL
    $119 + 3 = 122$ L 150 mL
13. **Correct Answer: B. 9 km 210 m**
    Distance covered by car
    Total distance: 20 km 560 m
    Distance by auto: 10 km 150 m
    Subtract: 20 km 560 m − 10 km 150 m = 10 km 410 m
    After walking **1 km 200 m**: 10 km 410 m − 1 km 200 m
    = 9 km 210 m
14. **Correct Answer: A. 103 kg 125 g**
    Total weight
    Arjun: 34 kg 375 g
    Kapil = twice Arjun's weight: 2 × 34 kg 375 g = 68 kg 750 g
    Total: 68 kg 750 g + 34 kg 375 g = 103 kg 125 g
15. **Correct Answer: B. 75 kg**
    Sold on Monday: $\frac{1}{5} \times 250 = 50$ kg
    Sold on Tuesday: $\frac{1}{10} \times 250 = 25$ kg
    Total sold: $50 + 25 = 75$ kg

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. **Correct Answers: A, C**

    Solution:
    The correct units to measure length are **Meter (A)**, **Centimeter (C)**, and **Millimeter (not listed, but part of the concept)**.
    Kilogram (B) is used to measure weight, and milliliter (D) is used to measure volume.
17. **Correct Answers: A, B**

    Solution:
    - 1 kilogram = 1000 grams (A)
    - 1 gram = 1000 milligrams (B)
    - 1 kilogram = 100 grams (C) is incorrect.
18. **Correct Answers: A, B**
    - 1000 milligrams = 1 gram (D) is incorrect.
      Solution:
    - **Area of rectangle = Length × Width (A)** is correct.
    - **Area of square = Side × Side (B)** is correct.
    - **Area of rectangle = 2 × (Length + Width) (C)** is incorrect, as this is the formula for perimeter, not area.
    - **Area of square = 4 × Side (D)** is incorrect, as this is the perimeter formula for a square.

19. **Correct Answers: A, C, D**

Solution:

- 1 liter = 1000 milliliters (A) is correct.
- 1 milliliter = 1000 liters (B) is incorrect; it should be the opposite.
- 500 milliliters = 0.5 liters (C) is correct.
- 1 cubic meter = 1000 liters (D) is correct.

20. **Correct Answers: A, B, D**

Solution:

- Perimeter of a square = 4 × Side (A) is correct.
- Perimeter of a rectangle = 2 × (Length + Width) (B) is correct.
- Perimeter of a square = Side × Side (C) is incorrect, as this formula is for area, not perimeter.
- Perimeter of a polygon is the sum of the lengths of its sides (D) is correct.

ASSERTION AND REASON TYPE

21. **Correct Answer: (A)**

Solution:

- The formula for the perimeter of a rectangle is indeed 2 × (Length + Width), and this formula correctly accounts for the total length around the rectangle.
- The reason correctly explains why the perimeter is calculated by adding all four sides, which match the formula.

22. **Correct Answer: (A)**

Solution:

- **Assertion (A)** is true as 1 kilometer is indeed equal to 1000 meters.
- **Reason (R)** is also true because kilometers are larger units than meters, which means 1 kilometer is equal to 1000 meters.
- The reasoning provided explains the assertion correctly.

23. **Correct Answer: (A)**

Solution:

- **Assertion (A)** is true because before performing any operations like addition or subtraction on lengths in different units, it is necessary to convert them into the same unit.
- **Reason (R)** is also true because adding or subtracting measurements in different units will lead to incorrect results unless the units are made uniform.
- The reasoning correctly explains the assertion.

MATRIX MATCHING TYPE

24. **Correct Answer:**
    - → (R) Kilograms
    - → (Q) Length × Width
    - → (P) Litter
    - → (S) 4 × side

Explanation:

- In the metric system, mass is measured in kilograms, but the imperial unit for mass is pounds.
- The area of a rectangle is calculated by multiplying the length by the width.
- The unit of volume in the metric system is the liter.
- The perimeter of a square is found by multiplying one side by 4

25. **Correct Answer:**
    - → (P) 1000 grams
    - → (Q) 2 × (Length + Width)
    - → (S) pound
    - → (R) Meter

Explanation:

- 1 kilogram is equal to 1000 grams.
- The perimeter of a rectangle is calculated by 2 × (Length + Width).
- The unit of mass in the imperial system is pounds
- The unit of length in the metric system is the meter.

COMPREHENSION TYPE

Passage-1

26. Correct Answer:

"Lucy's mom said they needed **500 grams of flour**."

So the correct answer is:

B) 500 grams

Explanation:
The answer is directly mentioned in the passage.

27. Correct Answer: C) 30 cm

"It was 30 centimetres long."

Explanation:
The measuring tape showed the baking pan length as 30 cm.

28. Correct Answer: C) 1 kilogram

"They found that the total weight was **1 kilogram**."

Explanation:
The cookies were weighed after baking, and their weight was 1 kilogram.

Passage-2

29. Correct Answer: B) 1000 cm^3

Solution:
Volume = $10 \times 10 \times 10 = 1000$

30. Correct Answer: A) 7200 cm^2

Solution:
Area = length × width
= $\mathbf{120 \times 60 = 7200}$

31. Correct Answer: B) 3

Solution:
Number of desks = $24000 \div 7200 =$ 3(approx.)

7. Geometry

Solutions

Worksheet – 1

1. **Correct Answer: A) Line**
   A line extends infinitely in both directions without any endpoints. A straight road going endlessly represents this property.
2. **Correct Answer: C) Line segment**
   A line segment has two fixed endpoints. A pencil also has two ends, so it matches a line segment.
3. **Correct Answer: B) Ray**
   A ray starts from one point and extends infinitely in one direction. Torch light starts from your hand (one point) and goes forward.
4. **Correct Answer: C) Semi-circle without closure**
   A semi-circle without closure does not form a complete boundary → open figure.
5. **Correct Answer: C) Made of both straight and curved lines**
   A closed figure can be made of straight lines, curved lines, or both—but it must form a boundary.
6. **Correct Answer: D**
   Option (D) forms a complete enclosed shape (arrow), while others are open.
7. **Correct Answer: B**
   Option (B) (star) is not completely closed, while others form closed shapes.

Worksheet – 2

1. **Correct Answer: B) Hexagon**
   A polygon with 6 sides is called a hexagon.
2. **Correct Answer: C) Circle**
   Polygons have only straight sides. A circle has curved edges, so it is not a polygon.
3. **Correct Answer: B) Concave polygon**
   An inward angle greater than 180° means the polygon is concave.
4. **Correct Answer: B) 540°**
   Sum = (n − 2) × 180
   = (5 − 2) × 180 = 540°
5. **Correct Answer: C) 6**
   Number of sides = 360 ÷ 60 = 6
6. **Correct Answer: B) 6**
   (n − 2) × 180 = 720
   n − 2 = 4 → n = 6
7. **Correct Answer: A) 52 cm**
   Add all the outer side lengths:
   10 cm + 4 cm + 5 cm + 5 cm + 14 cm + 14 cm
   = 52 cm

Worksheet – 3

1. **Correct Answer: (C) 180°**
   Sum of angles of a triangle
   A triangle always has interior angles adding up to 180°.
2. **Correct Answer: (C) Equilateral triangle**
   Triangle with all equal sides
   A triangle with all three sides equal is called an equilateral triangle.
3. **Correct Answer: (A) Twice the radius**
   Diameter of a circle
   The diameter is 2 × radius.
4. **Correct Answer: (A) 14 cm**
   Diameter when radius = 7 cm
   Diameter = 2 × 7 = 14 cm.
5. **Correct Answer: (B) 90°**
   Angle in ladder-wall-ground triangle
   Wall and ground are perpendicular, so one angle is 90°.
6. **Correct Answer: (B) Isosceles triangle**
   Type of triangle (5 cm, 5 cm, 8 cm)
   Two sides are equal (5 cm and 5 cm), so it is an isosceles triangle.
7. **Correct Answer: (A) Interior**
   Position of Point B
   Point B is inside the boundary of the circle, so it lies in the interior region.

Worksheet – 4

1. **Correct Answer: (B) A flat, 2D representation of a 3D shape**
   A net is when a 3D shape is opened out into a flat surface showing all its faces.
2. **Correct Answer: (B) Cube**
   Folding the given net
   The figure shows 6 equal squares arranged in a cross shape, which folds to form a cube.
3. **Correct Answer: (A) Two circles and one rectangle**
   Net of a cylinder
   A cylinder has two circular bases and one curved surface (rectangle when opened).
4. **Correct Answer: (C) Circle**
   Shape that cannot tessellate
   Triangles, squares, and hexagons tessellate, but a circle leaves gaps.
5. **Correct Answer: (B) Irregular tessellation**
   Different shapes fitting together without gaps (and not all equal) describe an irregular tessellation.
6. **Correct Answer: (C) Hexagon**
   Most commonly used tessellation shape
   Hexagons fit together perfectly without gaps (like honeycombs).
7. **Correct Answer: (A) Tiles on a floor**
   Real-life example of tessellation
   Tiles cover a floor without gaps or overlaps.

Worksheet – 5

1. **Correct Answer: (B) Circle**
   Figure with infinite lines of symmetry
   A circle has infinite lines of symmetry through its center.
2. **Correct Answer: (C) 4**
   Lines of symmetry in a square
   A square has 4 lines of symmetry (2 diagonals + 2 through midpoints).
3. **Correct Answer: (A) Scalene triangle**
   Shape with NO line of symmetry
   A scalene triangle has all sides different, so no symmetry line.
4. **Correct Answer: (D) 5**
   Lines of symmetry in a regular pentagon
   A regular pentagon has 5 lines of symmetry.
5. **Correct Answer: (B) 1**
   Lines of symmetry in a butterfly
   A butterfly has 1 vertical line of symmetry.
6. **Correct Answer: (C) A**
   Letter with mirror symmetry
   The letter A has a vertical line of symmetry, so it looks the same in a mirror.
7. **Correct Answer: (B) Right hand**
   Watch in mirror image
   A mirror reverses left and right, so the watch on the left hand appears on the right hand.

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (C) It extends infinitely in both directions**
   Characteristic of a line
   A line has no endpoints and goes on forever in both directions.
2. **Correct Answer: (B)**
   Example of a closed figure
   A closed figure has no gaps; the square is fully closed.
3. **Correct Answer: (B) 3**
   Red figure → Closed
   - Blue (spiral) → Open
   - Green → Closed
   - Light blue → Closed
   - Purple → Open

   Total closed figures = 3
4. **Correct Answer: (D)**
   (D) Incorrect (Diameter = 2 × Radius, so it is greater)
5. **Correct Answer: (B) Quadrilateral**
   A shape with 4 sides is a quadrilateral. Rectangle and square have equal opposite/all sides, but here all sides are different.

6. **Correct Answer: (D)**
   (D) Incorrect (Rectangle has opposite sides equal, not all different)
7. **Correct Answer: (C) III, II, I**
   Shape I: Perimeter = 2(10 + 7) = 34 cm
   - Shape II: Perimeter = 5 + 9 + 10 = 24 cm
   - Shape III: Perimeter = 5 + 3 + 7 + 7 = 22 cm

   Ascending order (smallest to largest):
   III (22) < II (24) < I (34)
8. **Correct Answer: (B) 82 cm**
   Horizontal sides
   Bottom = 19 cm
   Top segments = 4 + 5 + 4 + 6 = 19 cm
   Total horizontal = 19 + 19 = 38 cm
   Vertical sides
   Left + middle up = 8 + 14 = 22 cm
   Right + inner down = 12 + 4 = 16 cm
   Missing drop = 22 − 16 = 6 cm
   Total vertical = 8 + 14 + 4 + 12 + 6 = 44 cm
   Perimeter
   Perimeter = 38 + 44 = 82 cm
9. **Correct Answer: (C) Both A and B**
   All three sides are 5 cm, so all sides are equal.
   Hence, it is an equilateral triangle.
   Both (A) and (B) are correct.
10. **Correct Answer: (A) Square**
    A 4-sided figure with all sides equal is a square.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. **Correct Answer: (A), (C), (D)**

    Examples of polygons
    Polygons are closed figures with straight sides. Triangle, pentagon, and square fit this (circle does not).

12. **Correct Answer: (A), (B), (D)**

    Letters with vertical line symmetry
    A, M, and H can be divided into two equal halves vertically; B cannot.

13. **Correct Answer: (A), (B), (D)**

    Shapes that can tessellate
    Squares, triangles, and rectangles fit together without gaps; circles do not.

14. **Correct Answer: (A), (B), (C)**
    True statements about polygons and symmetry
    - Square → 4 lines of symmetry
    - Triangle (like equilateral) → has rotational symmetry
    - Hexagon (regular) → 6 lines of symmetry
    - Circle → has infinite symmetry
15. **Correct Answer: (A), (B), (C)**
    Shapes with both horizontal and vertical symmetry
    - Square
    - Rectangle
    - Circle
    - Triangle (only vertical in most cases)

ASSERTION AND REASON TYPE

16. **Correct Answer: (C)**

    Assertion (A): True — A line extends infinitely with no endpoints.
    Reason (R): False — This describes a line segment, not a line.

17. **Correct Answer: (C)**

    Assertion (A): True — A polygon is a closed figure made of straight-line segments.
    Reason (R): False — A circle is not a polygon (it has curved boundary).

18. **Correct Answer: (C)**

    Assertion (A): True — Tessellations cover a surface without gaps or overlaps.
    Reason (R): False — Circles do not tessellate; they leave gaps.

MATRIX MATCHING TYPE

19. **Correct Answer:**

    Match the following

    (a) A dot showing a location on a map → (Q) Point

    (b) A straight road extending endlessly in both directions → (P) Line

    (c) A rope with two fixed ends → (R) Line segment

    (d) A shape drawn without lifting pencil and ends meet → (S) Closed figure

20. **Correct Answer:**

Match the following

(a) A polygon with 3 sides → (P) Triangle

(b) A polygon with 4 sides → (Q) Quadrilateral

(c) A polygon with 5 sides → (S) Pentagon

(d) A polygon with 6 sides → (R) Hexagon

COMPREHENSION TYPE PASSAGE

21. **Correct Answer: (B) 180°**

Triangle angle sum
Sum of angles in a triangle = 180°.

22. **Correct Answer: (B) Radius**

Distance from center to edge
This distance is called the radius.

23. **Correct Answer: (C) Quadrilateral**

Shape with 4 sides
A shape with 4 sides is a quadrilateral.

Passage 2:

24. **Correct Answer: (B)**

Line symmetry
A figure that can be divided into two identical halves by a line.

25. **Correct Answer: (B) Two**

Lines of symmetry in a rectangle
A rectangle has 2 lines (horizontal and vertical).

26. **Correct Answer: (C) No, it does not have line symmetry**

Symmetry of a star
Given passage says a star does not have symmetry.

8. Data handling

Solutions

Worksheet – 1

1. **Correct Answer: C) 6**

   Given: 1 = 2 apples

   Number of stars = 3 stars

   So, total apples = 3 × 2 = 6

2. **Correct Answer: C) 15 apples**

   Solution: If 1 picture represents 5 apples, 3 pictures will represent 3×5=15 apples.

3. **Correct Answer: B) 40 books**

   Solution: If 1 symbol represents 10 books, 4 symbols represent 4×10=40 books.

4. **Correct Answer: D) 10 ice creams**

   Solution: Each star represents 2 ice creams, so 5×2=10 ice creams.

5. **Correct Answer: D) 18 pencils**

   Solution: Each picture represents 3 pencils, so 6×3=18 pencils.

6. **Correct Answer: B) 28 cars**

   Solution: Each picture represents 4 cars, so 7×4=28 cars.

7. **Correct Answer: C) 9 cupcakes**

   Solution: Since each picture represents 1 cupcake, and there are 9 pictures, the total number of cupcakes sold is 9.

Worksheet-2

1. **Correct Answer: B. To compare data visually**

   Solution: A bar graph is used to compare data using rectangular bars.

2. **Correct Answer: A. The number or value of the data it represents**

   Solution: The length of the bar corresponds to the quantity it represents.

3. **Correct Answer: B. X-axis**

   Solution: The horizontal line in a bar graph is the X-axis, where categories are usually labeled.

4. **Correct Answer: B. The unit used to measure data**

   Solution: The scale tells how much each unit of the bar represents, like 1 unit = 10 people.

5. **Correct Answer: C. 160 apples**

   Solution: If 1 cm = 20 apples, then 8 $cm = 8 \times 20 = 160\ apples$.

6. **Correct Answer: B. 45 books**

   Solution: If 1 cm = 5 books, then 9 $cm = 9 \times 5 = 45 books$.

7. **Correct Answer: C. 30 students**

   Solution: Since the bar for cricket is twice as tall, 15×2=3015 \times 2 = 3015×2=30 students like cricket.

Worksheet-3

1. **Correct Answer: A. Pie chart**

   Solution: A circle chart is commonly known as a pie chart because it looks like a pie divided into slices.

2. **Correct Answer: B. A part of the data or category**

   Solution: Each slice in a circle chart shows a portion of the total data.

3. **Correct Answer: B. One-fourth of the circle**

   Solution: 25% is one-fourth of the total circle.

4. **Correct Answer: B. 12.5%**

   Solution: Each slice represents $\frac{1}{8}$ of the total, which equals 12.5%.

5. **Correct Answer: A. Half of the students like mango the most.**

   Solution: 50% indicates half of the total students prefer mango.

6. **Correct Answer: C. 33.33%**

   Solution: A full circle is 360°. Food expenses covering 120° means $\frac{120^{\circ}}{360^{\circ}} \times 100 = 33.33\%$

7. **Correct Answer: D. 4 slices**

   Solution: A full circle is 360°, so $\frac{360^{0}}{90^{0}} = 4$. Four slices will fit.

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: C) Pictures or symbols**

   Solution: A pictograph visually represents data using images or symbols to show quantities, making it easier to understand.

2. **Correct Answer: C) The value or frequency of that category**

   Solution: The height of each bar in a bar graph shows the quantity or frequency of the category it represents.

3. **Correct Answer: C) To ensure accurate representation of data**

   Solution: Scaling is important in bar graphs to ensure that data is represented proportionally and accurately.

4. **Correct Answer: C) Circle chart (pie chart)**

   Solution: A circle chart (pie chart) effectively shows how different parts contribute to a whole, making it ideal for illustrating proportions.

5. **Correct Answer: A) 90 degrees**

   Solution: Since a circle is 360 degrees, 25% of 360 degrees is 90 degrees (25/100 × 360).

6. **Correct Answer: A. 5**

   How many less students participated from Section B than Section C and D together?

   Students in **C and D together**
   = 9 + 4
   = **13**

   Students in **B** = 8

   Difference = 13 – 8 = **5**

7. **Correct Answer: C. 5/31**

   Fraction of students from Section E to total students

   Students in **E** = 5

   Total students
   = 5 + 8 + 9 + 4 + 5
   = **31**

   Required fraction
   = 5 / 31

8. **Correct Answer: 1000**

   (i) On which day were maximum packets distributed?

   The highest number is on **Wednesday = 3500**

   (ii) How many more packets were distributed on Thursday and Friday together than on Tuesday?

   Thursday + Friday
   = 2000 + 1500
   = **3500**

   Tuesday = **2500**

   Difference = 3500 – 2500 = **1000**

9. **Correct Answer: C. 24**

   Total number of projects completed by Group M and Group T

   Group M + Group T
   = 9 + 15
   = **24**

10. **Correct Answer: A. 1/2**

    Fraction of projects completed by Group W to Group J

    Group W = **6**
    Group J = **12**

    Fraction = 6 / 12 = **1/2**

11. **Correct Answer:**

    Find the difference between Varun and Karan

    Varun = **90 cm**
    Karan = **75 cm**

    Difference = 90 – 75

$$90 - 75 = 15$$

Difference = 15 cm

12. Correct Answer: 350

(a) In which year did the newspaper get maximum subscribers?

The highest number is **400** in **2016**

(b) How many less subscribers in 2012 than 2014 and 2015 together?

2014 + 2015
= 350 + 250
= **600**

2012 = **250**

Difference = 600 – 250

$$600 - 250 = 350$$

13. Correct Answer: D. Rupali and Arun

D. Rupali and Arun

= 65 + 60 = **125**

$$65 + 60 = 125$$

14. Correct Answer: Kesar

(a) Which milk flavour is liked by exactly 40 students?

From the graph, **Kesar = 40**

(b) How many students like Rose, Badam and Pista altogether?

Rose + Badam + Pista
= 20 + 50 + 30

$$20 + 50 + 30 = 100$$

Total = 100 students

15. Correct Answer:

Find how many fewer scooters than cars

Difference = Cars – Scooters
= 80 – 40

$$80 - 40 = 40$$

Difference = 40

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. Correct Answer: (A) Uses symbols or pictures to represent data.
(C) Requires a key to understand the values.

Explanation: A pictograph uses symbols to represent data, and a key explains what each symbol represents. Bars are used in bar graphs, not pictographs. Showing parts of a whole is done in pie charts.

17. Correct Answer: (A) Easy to read and understand.
(C) Allows quick comparison of data.
(D) Requires minimal numerical calculations.

Explanation: A pictograph simplifies data representation and comparison but always requires a key to interpret data.

18. Correct Answer: (A) Use a consistent scale.
(B) Draw bars of equal width.
(D) Label the axes and provide a title.

Explanation: A bar graph requires a consistent scale, equal-width bars, and proper labeling for clarity. A separate graph for each category is not necessary.

19. Correct Answer: (A) The length of the bars represents the data value.
(B) Bars can be horizontal or vertical.
(D) A scale is always required.

Explanation: Bar graphs represent data using the height or length of the bars, which can be horizontal or vertical. The space between bars does not carry data meaning, but a scale is necessary for accurate interpretation.

20. Correct Answer: (A) Calculate the percentage for each category.
(C) Divide the circle based on percentages.

Explanation: A pie chart involves calculating percentages for each category and dividing the circle accordingly. Scales and bars are used in bar graphs, not pie charts.

ASSERTION AND REASON TYPE

21. Correct Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

Explanation: A bar graph compares data by using the height or length of bars to represent values.

22. **Correct Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).**

    Explanation: A pie chart divides data into slices representing percentages, showing how each part relates to the whole.

23. **Correct Answer: (D) Assertion (A) is false, and Reason (R) is true.**

    Explanation: Pictographs are simple but not suitable for very large data as scaling symbols can make them hard to interpret.

MATRIX MATCHING TYPE

24. **Correct Answer:** $a \rightarrow Q, b \rightarrow P, c \rightarrow R, d \rightarrow S$

    (a) Pictograph uses pictures or symbols to represent data.

    (b) Bar Graph represents data using bars of different lengths.

    (c) Circle Chart represents data as parts of a whole circle.

    (d) Scaling ensures consistent representation of data values.

25. **Correct Answer:** $a \rightarrow S, b \rightarrow Q, c \rightarrow R, d \rightarrow P$

    - Circle Chart is best for showing proportions.
    - Bar Graph is used to compare data categories.
    - Pictograph visually represents data.
    - Scaling uses equal intervals.

COMPREHENSION TYPE PASSAGE

26. **Correct Answer: B) 20**

    Each symbol = 5 students
    Long Jump has 4 symbols
    4 × 5 = 20

27. **Correct Answer: C) RELAY**

    Running = **30**
    Long Jump = **20**
    Relay = **45**
    Greatest number = **45**

28. **Correct Answer: B) 30**

    Current Long Jump = **20**
    A better target from options is **30**

PASAAGE -2

29. **Correct Answer: B) 15**

    How many oranges were sold compared to grapes?
    Oranges = 20
    Grapes = 5
    Difference = 20 – 5 = 15

30. **Correct Answer: C) GRAPES**

    Which fruit had the least sales?
    Apples = **15**
    Bananas = **10**
    Oranges = **20**
    Grapes = **5**
    Strawberries = **12**
    Smallest number = **5**

31. **Correct Answer: C) 62**

    What is the total number of fruits sold?
    15 + 10 + 20 + 5 + 12 = 62